

Kestrel Operations Handbook v4.2 (2026)

This handbook is binding for all flight operations staff. It replaces version 4.1 from November 2025.

1. Fleet overview

Kestrel operates three drone models. Payload limits drop in rain because wet propellers lose lift; the rain limit applies whenever precipitation is reported at the launch or drop site.

Model	Max payload (dry)	Max payload (rain)	Range	Max wind	Battery swap
K-2 Sparrow	1.2 kg	0.9 kg	12 km	28 km/h	every 40 cycles
K-4 Kestrel	2.5 kg	1.8 kg	18 km	35 km/h	every 60 cycles
K-7 Heron	6.0 kg	4.2 kg	25 km	42 km/h	every 80 cycles

2. Weather rules

No flight may start when the forecast wind at flight altitude exceeds the model's maximum wind rating. Lightning: if a strike is detected within 15 km of the route, all drones on that route are grounded until 30 minutes after the last strike (the 15/30 rule). Below -10 degrees Celsius batteries must be pre-warmed to at least 15 degrees before launch.

3. Night operations

Night flights (sunset to sunrise) require form NW-17, signed by the shift lead before the first launch of the night. The maximum altitude at night is 90 m above ground; during the day it is 120 m.

4. Battery policy

A battery whose state of health falls below 82 % is retired and sent to recycling. Batteries in storage are kept at a 55 % charge. Every battery carries a QR label with its cycle count.

5. Incident codes and escalation

Code	Meaning	Escalate to	Within
KX-1	Minor deviation, no damage	Shift lead	end of shift
KX-2	Payload lost or damaged	Operations manager + customer service	1 hour
KX-3	Injury or third-party damage	Managing director + authority (LBA)	15 minutes